

HDOC Day-15

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases. Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice: Check whether all digits are even. E.g. 1: Number: 12345, Output: All digits not even E.g. 2: Number: 24680, Output: All digits are even

1. Algorithm

1. Start.
2. Display the menu:
 - 1. Check whether all digits are even
 - 2. Exit
3. Enter the user's choice.
4. Use a switch statement.
5. If the choice is 1:
 - Input a positive integer num.
 - Set temp = num.
 - Set allEven = 1.
 - Repeat while temp > 0:
 - Find the last digit using digit = temp % 10.
 - Check digit % 2 != 0.
 - If true, set allEven = 0 and stop checking.
 - Remove the last digit using temp = temp / 10.
 - If allEven == 1, display **"All digits are even"**.
 - Otherwise, display **"All digits are not even"**.
6. If the choice is 2, exit the program.
7. For any other choice, display **"Invalid choice"**.
8. Stop.

3. Test case table

Test Case	Choice	Input Number	Expected Output	Actual Output	Result
1	1	24680	All digits are even	All digits are even	Pass
2	1	12345	All digits are not even	All digits are not even	Pass
3	1	8642	All digits are even	All digits are even	Pass
4	1	24681	All digits are not even	All digits are not even	Pass
5	1	22222	All digits are even	All digits are even	Pass

4.C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int choice, num, temp, digit, allEven;
```

```
    printf("1. Check whether all digits are even\n");
```

```
    printf("2. Exit\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    switch (choice)
```

```
{
```

```
    case 1:
```

```
        printf("Enter a positive integer: ");
```

```
        scanf("%d", &num);
```

```
        if (num <= 0)
```

```
{  
    printf("Please enter a positive integer.\n");  
    break;  
}
```

```
temp = num;  
allEven = 1;
```

```
while (temp > 0)  
{  
    digit = temp % 10;
```

```
    if (digit % 2 != 0)
```

```
    {  
        allEven = 0;  
        break;  
    }
```

```
    temp = temp / 10;  
}
```

```
if (allEven == 1)  
    printf("All digits are even\n");  
else  
    printf("All digits are not even\n");
```

```
break;
```

case 2:

```
printf("Exiting program...\n");
```

```
break;
```

default:

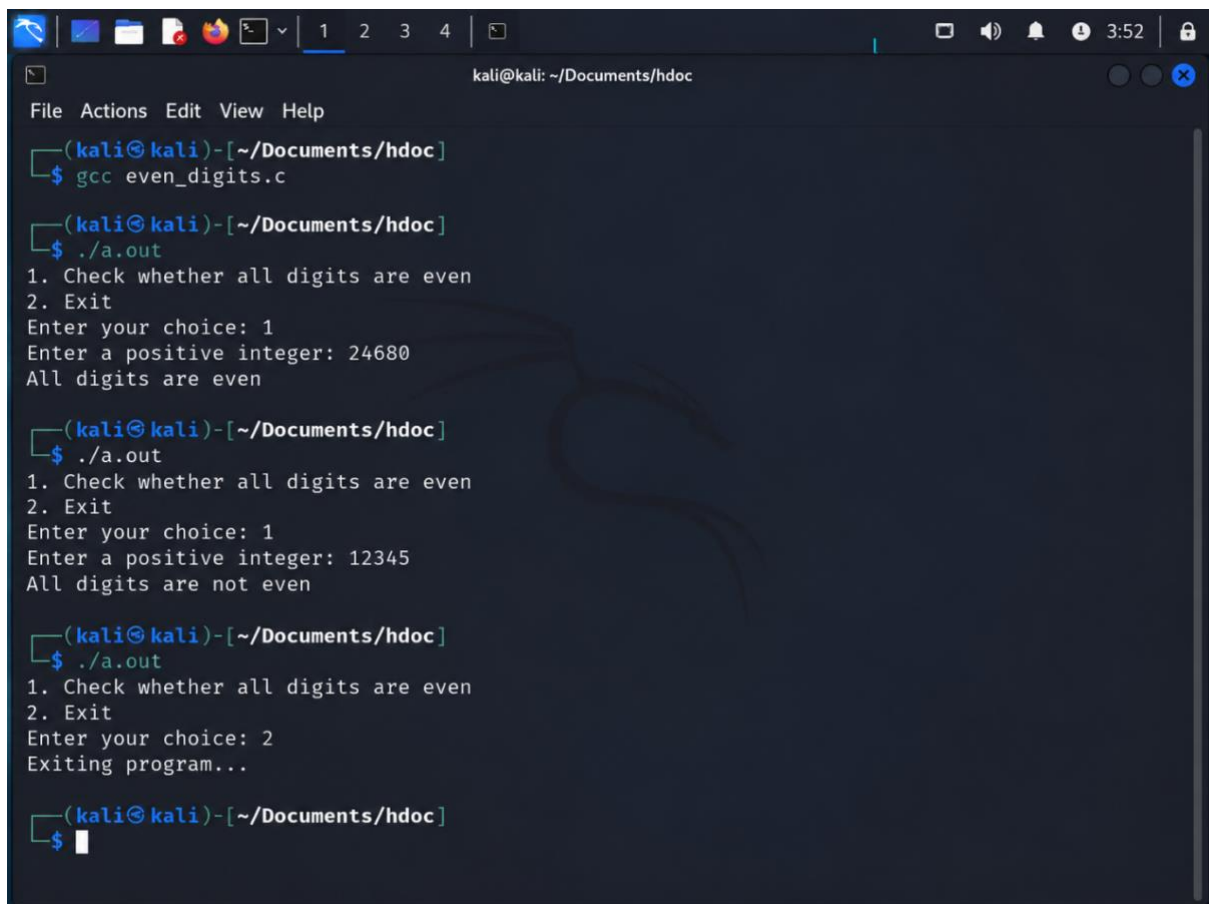
```
printf("Invalid choice\n");
```

```
}
```

```
return 0;
```

```
}
```

5. Output



```
kali@kali: ~/Documents/hdoc
File Actions Edit View Help
(kali@kali)-[~/Documents/hdoc]
$ gcc even_digits.c
(kali@kali)-[~/Documents/hdoc]
$ ./a.out
1. Check whether all digits are even
2. Exit
Enter your choice: 1
Enter a positive integer: 24680
All digits are even
(kali@kali)-[~/Documents/hdoc]
$ ./a.out
1. Check whether all digits are even
2. Exit
Enter your choice: 1
Enter a positive integer: 12345
All digits are not even
(kali@kali)-[~/Documents/hdoc]
$ ./a.out
1. Check whether all digits are even
2. Exit
Enter your choice: 2
Exiting program...
(kali@kali)-[~/Documents/hdoc]
$
```

The program correctly checks every digit and produces the expected results for the prepared test cases.